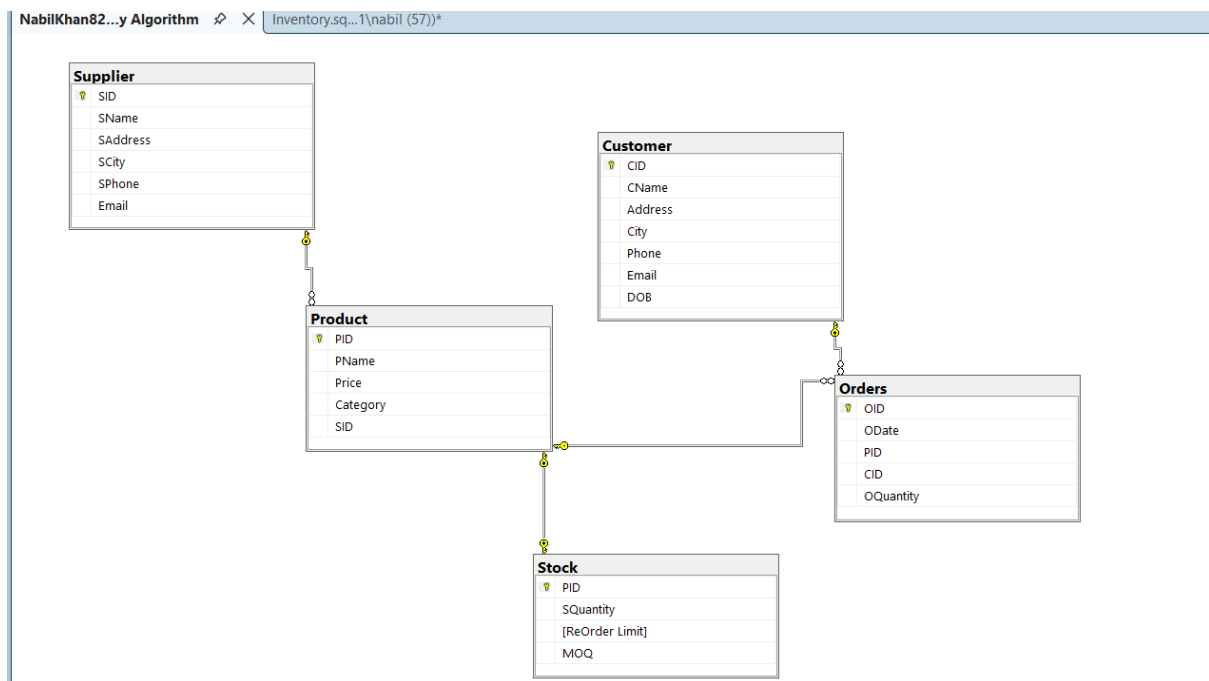


Inventory Management System

(SQL Project)

This project manages suppliers, products, stock, customers, and orders. It helps analyse sales data, track inventory, and identify business insights using SQL.



As you can see, the database contains 5 tables in total.

They are linked with each other through Primary Key and Foreign Key constraints.

1. Supplier has 6 fields: SID (PK), Sname, SAddress (CK), SCity (CK), SPhone, Email.
2. Product has 5 fields: PID (PK), PName, Price (CK), Category (CK), SID (FK).
3. Stock has 4 fields: PID (FK) (PK), SQuantity (CK), ReOrder Limit (CK), MOQ (Minimum Order Quantity) (CK).

4. Customer has 7 fields: CID (PK), CName, Address, City, Phone (UQ), Email (UQ), DOB (CK).
5. Orders has 5 fields: OID (PK), ODate, PID (FK), CID (FK), OQuantity (CK).

Key Features:-

1. Track stock levels
2. Identify low stock
3. Calculate revenue
4. Analyse customer purchases
5. Displays all tables

Queries:-

1. Product & Supplier Details

```
select PID, PName, Category, SName, SCity  
from Product  
left join Supplier  
on Product.SID = Supplier.SID
```

 *Displays each product along with its supplier information.*

2. Order Summary

```
select OID, ODate, CName, Address, Phone, PName, Price,  
OQuantity,  
Price * OQuantity as Revenue  
from Orders  
inner join Customer  
on Customer.CID = Orders.CID
```

inner join Product

on Product.PID = Orders.PID

👉 *Shows complete order details with calculated revenue for each order.*

3. Revenue per Customer

select Customer.CID, CName,

coalesce(sum(Price * OQuantity), 0) as Revenue

from Customer

left join Orders

on Orders.CID = Customer.CID

left join Product

on Product.PID = Orders.PID

group by Customer.CID, CName

order by Revenue desc

👉 *Calculates total revenue generated by each customer.*

4. Top 3 Products by Quantity Sold

select top 3 Product.PID, PName, OQuantity as [Highest Quantity]

from Product

inner join Orders

on Orders.PID = Product.PID

order by [Highest Quantity] desc

👉 *Identifies the top 3 products based on highest quantity sold in a single order.*

5. Low Stock Products

```
select Product.PID, PName, SQuantity, [ReOrder Limit]
from Product
inner join Stock
on Stock.PID = Product.PID
where SQuantity < [ReOrder Limit]
order by SQuantity asc
```

👉 *Finds products that need restocking based on reorder limits.*

6. Products Never Sold

```
select Product.PID, PName, OQuantity
from Product
left join Orders
on Product.PID = Orders.PID
where OQuantity is null
```

👉 *Lists products that have never been ordered.*

7. Revenue by Category

```
select Category, coalesce(sum(Price * OQuantity), 0) as
Revenue
from Product
left join Orders
on Orders.PID = Product.PID
group by Category
order by Revenue desc
```

👉 *Shows total revenue generated for each product category.*

Conclusion:-

This project demonstrates the use of SQL to analyze business data through a relational database structure. It involves multiple tables, such as Product, Customer, Orders, Supplier, and Stock, connected using keys and constraints. Key SQL concepts applied include joins, aggregation functions, filtering, and handling null values. The queries solve real-world problems like tracking revenue, identifying best-selling products, managing inventory, and analyzing customer contributions. This project strengthened practical SQL skills and improved understanding of data-driven decision-making.